A Predictive Analysis of the 2020 Presidential Election Outcomes Using Advanced Modelling Techniques

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Introduction This following report aims to delve into the landscape of contemporary politics in the United States of America. The focus is brought to the presidential elections and our ability to accurately predict voting outcomes through the use of modelling techniques. Utilizing a diverse set of variables provided or a curated subset, each depicting the multifaceted demographic of the United States, our objective is to develop a robust model capable of predicting the outcome of the "majority" variable.

The investigation is composed of these principal objectives:

1) Construct a model which accurately predicts the outcome of the votes depicted by the "majority" variable using the other variables, or a subset of these. To concisely explain the rationale behind variable selection.
2) Demonstrate the efficiency of the model by splitting the data into training and tests data and through the employment of rigorous testing procedures.
3) Furnish the bank with an insightful and actionable tool for navigating the complex terrain of electoral predictions by means of a comprehensive report with valuable findings.

Denoted by either "Trump" or "Biden", the "majority" variable encapsulates much more than the most popular candidate, it reflects the intricate diversity of preferences within the United States electorate. A myriad of variables help paint the picture of the nation's political landscape, ranging from socioeconomic factors to regional characteristics. By understanding the importance of these variables we can develop a model that discerns patterns and relationships, ultimately predicting the outcome of votes in the 2020 election.

Preliminary Analysis Prior to developing a predictive model, an initial exploration of the dataset was conducted to yield a nuanced understanding of the structure and each variable. Upon loading the dataset "data_election_2020.Rdata", undesirable missing values were identified. To mitigate bias and distortion and to ensure a representative analysis, missing values were handled by mean imputation and row deletion. The inclusion of both (not simultaneously) was to mitigate the assumptions made by mean imputation and to preserve the size of the data affected by row deletion.

Upon a comprehensive exploration of the electoral data, a clear pattern emerges, one which underscores the electoral strength of Democratic candidates over the past elections. In fact, Barack Obama secured a substantial victory over Mitt Romney in 2012, amassing a noteworthy margin of 4,639,044 votes. The trend persisted in 2016 when Hillary Clinton outpaced Donald Trump by holding 48% of the votes, solidifying the Democrats' consistent strength in the popular vote. On closer examination of the U.S. electoral system and the dataset overall, it was uncovered that Trump accounts for a large tally in the "majority" variable due to his success in winning over many low-density counties. However, the counties that Biden dominates are more often than not counties that form the economic backbone of the United States, encompassing "71% of America's economic activity"(Muro et. al, 2020). The dataset displays significant variation in the number of counties across states, thus skewing the influence of states with a higher number of counties yet lower populations within. This discrepancy affects the relationship between variables and impairs our ability to curate a balanced set of predictors. Recognizing the need for a more intricate approach, I aimed to address class imbalance in the "majority" variable. To create a model that could reflect this geopolitical nature of the United States these are the three methods I chose:

1) Stratification: strata were created based on both "state" and "county". In doing so, county-level dynamics within each state are accounted for, as well as regional variations in voting patterns.
2) Resampling: by creating weights reflecting the proportion of votes each candidate amassed in relation to an accumulative population density of the counties the data was resampled. Through this equitable approach the minority class (majority == "Biden") was increased.
3) State Levelled Data: similar to stratification above, this method aggregates each county to its respective state and summarises all its statistics by calculating totals and mean values for each variable on a state level.

Analysis By establishing a correlation matrix between various variables, I wanted to unravel any interdependencies among demographic factors, socio-economic indicators, and voting outcomes. Using the dataset altered on a state level, as it showed the most meaningful correlations, I produced a correlogram. However, with almost 22 variables to analyse, the correlogram becomes cumbersome to pick apart. In hopes of extracting a meaningful assortment of predictors for my model, I took the approach of building voter profiles. This included looking at individual correlations between descriptors of voter demographics and their relationship to candidates. Below are two summary profiles which focus on the following attributes: demographics, education, age, income and rural vs. urban influence.

The Conservative Leaning Electorate Profile: In regions characterized by a higher proportion of white residents, lower educational attainment, an aging demographic, reduced median household incomes, and predominantly rural settings, there is a discernible inclination for voters to align with Republican candidates.

	white_pct	black_pct	hispanic_pct	rural_pct	median_hh_income
Trump_16	-0.16	0.12	0.27	-0.17	-0.11
Clinton_16	-0.2	0.01	0.34	-0.33	0.12
0.89	age29andunder	age65andolder	clf_unemployment	lesscollege_pct	lesshs_pct
Trump_16	0.09	-0.06	0.33	0.2	0.37
Clinton_16	0.05	-0.1	0.29	-0.05	0.2

Figure 1: Sample of correlations for Republican Candidates

The Liberal Leaning Electorate Profile: Counties with elevated percentages of Black and Hispanic populations, a higher prevalence of college-educated individuals, a younger demographic, increased median household incomes and urban landscapes exhibit a pronounced tendency to support Democratic candidates.

	white_pct	black_pct	hispanic_pct	rural_pct	median_hh_income
obama_12	-0.16	0.01	0.29	-0.32	0.11
romney_12	-0.19	0.11	0.30	-0.18	-0.10
0.91	age29andunder	age65andolder	clf_unemployment	lesscollege_pct	lesshs_pct
obama_12	0.02	-0.09	0.29	-0.04	0.17
romney_12	0.13	-0.91	0.32	0.19	0.38

Figure 2: Sample of correlations for Democratic Candidates

From creating these profiles, it became clear to me that the set of predictors should be able to encapsulate the nuances in every type of individual represented by the data and their tendencies towards any particular party or candidate. The chosen predictors shown below were used in the optimal model. They capture both electorate profiles well and in such a way that multicollinearity with other potential predictors are mitigated, overall minimising the amount of predictors needed to effectively predict the "majority" variable.

```r
selected_predictors <- c(
  "average_age65andolder_pct", "average_lesscollege_pct",
  "average_lesscollege_whites_pct", "average_rural_pct",
```

```
  "average_foreignborn_pct", "average_white_pct","average_clf_unemploy_pct","median_hh_inc"
)
```

Creating a Generalized Linear Model (GLM) Due to the binary nature of the "majority" variable, the generalized linear model chosen for this analysis was the logistic regression model. The chosen GLM allows for a nuanced assessment of the impact of each predictor variable on the odds of the "majority" outcome which would allow us to modify and enhance our model. The decision to not standardize the dataset despite certain variation in scales in the numerical variables (Percentages vs. Counts) was approached to preserve the interpretability of GLM coefficients. Due to working with an amalgamated dataset, it cannot be said that multicollinearity was at a minimum. The decision allows the coefficients of the logistic regression model to directly reflect the change in the log-odds of the outcome associated with a one-unit change in the predictor variable.

An exhaustive number of models were constructed, each on a different dataset using 2 different techniques of data handling. First on the original data which served as a control, stratified data, state levelled data and finally on resampled data. However, upon background testing, I found that the performance of the model built upon stratified data was the exact same for the model trained on the original dataset. This led me to believe my attempt at stratification failed. Regardless, it was kept for comparative purposes and is now named "control". Below is a table of many of the models tried and tested.

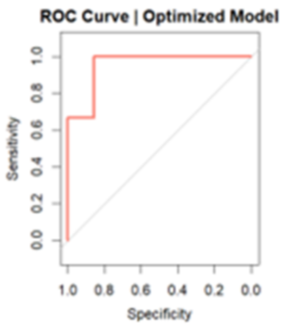
Type of Data	State Level Subsetted Data	-	-	-
Reproducibility	Seed (21)	Seed (21)	Seed (21)	Seed (21)
Data Handling Method	Mean imputation	Mean imputation	Deleting rows	Deleting rows
Predictors	all variables	less variables	all variables	less variables
AUC	AUC-ROC: 0.9285714	AUC-ROC: 0.7142857	AUC-ROC: 0.8	N/A
Accuracy	Accuracy : 0.9	Accuracy : 0.7	Accuracy: 0.667	N/A
Type of Data	Control	Resampled Data	Iteration #2	
Reproducibility	Seed (220307)	Seed (220307)	Seed (220307)	
Data Handling Method	Mean Imputation	Mean Imputation	Mean Imputation	
Predictors	All variables	All variables	less variables	
AUC	AUC-ROC: 0.8892817	AUC-ROC: 0.926	AUC-ROC: 0.912	
Accuracy	Accuracy: 0.9455446	Accuracy: 0.906	Accuracy: 0.8943	
Type of Data	Control	Resampled Data	Iteration #2	
Reproducibility	Seed (220307)	Seed (220307)	Seed (220307)	
Data Handling Method	Row Deletion	Row Deletion	Row Deletion	
Predictors	All variables	All variables	less variables	
AUC	AUC-ROC: 0.9973304	AUC-ROC: 0.909279	AUC-ROC: 0.9153036	
Accuracy	Accuracy: 0.9840849	Accuracy: 0.8514589	Accuracy: 0.8567639	

Figure 3: Table of tried and tested models

For each model, they were trained and tested utilizing an 80/20 split and initially using all the variables as the predictors. The 80/20 split allowed a pragmatic balance between giving the model substantial access to data for training in order to ensure model complexity and accuracy, and reserving a portion for testing, which was crucial in assessing the model's performance. Through trial-and-error efforts, 70/30 and 90/10 ratios were also attempted but were abandoned due to unparalleled accuracy.

Then, a comprehensive evaluation was conducted by means of performance metrics, such as accuracy and precision scores, which provide a detailed breakdown into the model's ability to correctly classify instances and finally examining Receiver Operating Characteristic (ROC) curves. By comparing each model on this evaluation rubric, the model built on the state levelled data was most optimal. However, the model on resampled data and using the same parameters performed very closely in comparison. While the former model (state levelled model) becomes our focus in the report, by aggregating individual counties to their state and subsequently their respective values for other variables, the data I was working on became significantly smaller and this was a factor to be considered.

The Optimal Model: State Levelled Data, Mean Imputation, Selected Predictors Taking a look at the performance metrics below, we can evaluate that the model was able to correctly classify outcomes in the test set, true positives and false negatives with 90% accuracy. The perfect precision of 100% signifies that when the model predicts a particular electoral outcome, it does so without false positives, enhancing confidence in the accuracy of positive predictions.However, the recall of 0.75 suggests that the model may miss a quarter of actual positive outcomes, potentially indicating a limitation in capturing all nuances of voter preferences. For high-stakes elections such as the 2020 Presidential Election, the consequences of mispredictions can be significant, making this trade off in the recall score costly.



Predictor	Log Odds
Intercept	1282.93
age65andolder	-11.90
avg_clf_unemployment	-23.05
median_hh_inc	0.00
avg_lesscollege	-46.86
avg_lesscollege_whites	38.88
avg_rural	1.79
avg_foreignborn	17.08
avg_white	-6.31

AUC	0.952381

Accuracy	0.90
Precision	1.00
Recall	0.75
F1 Score	0.86

Figure 4: Performance Metrics, ROC Curve and AUC Score

Turning to the ROC curve , we can tell from the shape of the graph its efficacy in classifying the voting outcomes. The point on the ROC curve corresponds to a specific threshold. The closer the point is to the top-left corner of the plot, the better the model, as it indicates the overall classification ability of the model. These curves also proved to be a helpful tool in selecting appropriate threshold values, for all models a threshold value of 0.7 was used. The Area Under the Curve (AUC) value indicates how effective a model is in distinguishing between classes. The optimal model has a value of 0.952381, indicating that the model is 95.24% successful at classifying the outcome of the "majority" variable. This high AUC score reflects the model's excellent capability in distinguishing between counties or states where a specific voting outcome occurred and where it did not. This indication of strong predictive power suggests that the model is taking to the predictors well and is able to capture underlying patterns.

The GLM outputs data depicting coefficients and the log odds associated with each predictor, allowing us to determine the log-odds of classifying the event "Biden_Win" (where majority == "Biden"). The data was transformed into the more readable table shown above. Positive values with larger magnitudes indicate a greater likelihood that a given variable is associated with the outcome of Biden winning the election in a particular state. The log-odds denote a relationship whereby a 1 unit increase in a predictor variable is associated with an increase/decrease in the log-odds and therefore an increase/decrease in the probability of the event. Looking at "average_age65andolder_pct", a one-unit increase in the percentage of the population aged 65 and older is associated with a decrease in the log-odds by 11.90. This stands true to the electorate profiles created earlier in our analysis, that senile demographics have a tendency to lean away from Democratic parties and instead, towards Republican candidates. Similarly, an increase in "average_lesscollege_pct" shows the probability in classifying as "Biden" decreases by 46.86, suggesting that those who are less educated may stray away from voting for Biden who is widely seen as the more progressive electoral candidate in 2020. As the log odd probabilities in classifying the outcome for "majority" is in line with our exploratory correlation work, we can be confident in our selection of predictors for the model. However, we can notice that "median_hh_inc" has a log odd of 0, rendering it insignificant for the model and therefore can be removed to potentially enhance the model.

Model Refinement Through the removal of predictor variables associated with low log odds, we sought to refine the model such that the AUC value cannot be enhanced further. However, by removing "median_hh_inc", I was able to yield an AUC of 1, or 100% classification rate. Regardless of this enhancement, this score unassumingly highlights the curse of perfect separation, whereby we have curated a combination of predictors that perfectly predicts the outcome variable. Due to convergence errors received by running this improved model, I thought of this enhancement to be negligible for the sake of preserving overall accuracy.

Final Discussion As mentioned at the beginning of this report, the primary objective was to "Construct a model which accurately predicts the outcome of the votes depicted by the"majority" variable using the other variables, or a subset of these".

Through my analysis, undertaking generalised linear models, more specifically a logistic regression model, I was able to successfully predict the outcome of the votes in the 2020 presidential election based on a particularly devised set of predictor variables with 95% accuracy.

Despite this accuracy in predicting, the techniques that enabled me to reach this point only further elucidated the nature of the American electoral system. While the dataset conveys information detailing the influence behind popular votes, it prompted a crucial consideration of the broader context, particularly the significance in electoral votes which ultimately dictate the winner of the presidential election. I began to ask; how much predictive power is in the hands of an ordinary individual as depicted by the dataset? At the end of the day, each voter acts in their best interest and in response to a melody of economic and political factors that affect their day-to-day life. On the other hand, the governance structure of the United States, encompassing 50 states, reflects the diverse array of needs and wants within a vast population too. This additional layer of intricacies within each state and even further, within each county, underscores that predicting a binary outcome such as the "majority" variable isn't as simple as it seems.

While the data reveals interesting patterns, it is crucial to understand that these findings serve as a basis for predicting candidate popularity. To improve the model's predictive power, the limitations of the dataset, especially its outdated nature (data only available for 2012-2016) need to be revisited, with an emphasis for a more current and comprehensive representation. Considering the dynamic political landscapes and the impact of recent events such as the Coronavirus pandemic and Roe v. Wade and the subsequent responses of political parties to these events, updating the dataset with more recent information could enhance the predictive power of the model for a wide range of uses, both economical and political. In conclusion, this report acts as a valuable starting point for comprehending the predictive nature of electoral dynamics. Against the backdrop of the geopolitical powerhouse that is the United States, I have highlighted the necessity of continuously updating the input that fuels predictive models as to account for fluctuations in demographics and their voting decisions in response to current events.

References Muro et. al. (2020). Biden-voting counties equal 70% of America's economy. What does this mean for the nation's political-economic divide? Brookings. Available at: https://www.brookings.edu/articles/biden-voting-counties-equal-70-of-americas-economy-what-does-this-mean-for-the-nations-political-economic-divide/ [Accessed 29 Nov. 2023]